

Audio-Visual Approximation of Video Semantic Space

Yasir Farooqui

Department of Computer Engineering,
Faculty of Engineering and Technology,
Jamia Millia Islamia, New Delhi

Abstract

Dense temporal video processing presents a severe computational bottleneck in modern multimodal representation learning workflows. This work investigates the feasibility of an alternative approach: the Audio-Visual Approximation of Video Semantic Space.

The primary objective is to determine whether the semantic embeddings of an entire video sequence can be accurately approximated using highly sparse, partial multimodal inputs—specifically, a single static visual frame paired with its corresponding audio track.

The project outlines an end-to-end framework designed to extract independent audio and visual features and map them into a shared latent space that mirrors ground-truth full-video embeddings.

Multiple representation learning methodologies including contrastive alignment (InfoNCE loss), latent space regression, cross-modal distillation, and Transformer-based fusion are explored. The resulting frameworks are evaluated in three dimensions: latent proximity, downstream retrieval performance (e.g. $R@K$, Median Rank), and computational efficiency.

The goal is to deliver competitive semantic retrieval performance while significantly reducing the computational cost and inference latency inherent to dense video modeling.

1 Introduction

Audio-Visual Representation Learning frameworks fundamentally depend on processing dense frame sequences across space and time. State-of-the-art multimodal vision models routinely ingest large quantities of consecutive video frames to map high-level action boundaries and semantic trajectories. However, dense temporal processing represents an acute computational bottleneck during large-scale repository indexing, real-time downstream retrieval, and edge inference. The heavy reliance on temporal convolutions, 3D space-time kernels, and multi-frame video self-attention creates crucial latency bottlenecks.

To address these limitations, the project investigates a lightweight paradigm: the *Audio-Visual Approximation of Video Semantic Space*. It implements an elegant structural compression method in which the complete temporal context of a full video clip is compressed into highly sparse, sub-sampled multi-modal streams—specifically, a single visual

keyframe (or sparse static visual descriptors) paired with its corresponding low-rate audio track. The missing visual temporal trajectory is compensated for by the sequential density and context of the native audio track.

We construct a decoupled architecture that fuses disparate acoustic and visual features into a unified latent space. Our framework maps these low-overhead proxy representations directly against target ground-truth full-video embeddings, delivering near-parity downstream retrieval and class metrics at a minute fraction of the computational and inference cost.

2 Dataset

Our model is an end-to-end ML pipeline which can be introduced to any Audio-Visual dataset. For the sake of experimentation and evaluation, we use pre-extracted representations from the **AVMIT (Audio-Visual Moments in Time)** dataset (1), packaged via standardized gzipped `.tfrecord` sequence structures. The primary dataset pipeline skips costly raw video decoding and heavy frame-rate resizing operations by working directly inside a specialized feature space extracted via standard foundation frameworks.

Each sequence sample within the dataset provides three core attributes extracted globally across the temporal extent of a video sequence:

1. **Visual Embeddings:** Sourced via a pre-trained VGG16 network architecture (3) from discrete keyframes. Represented as a sequence of shape $(T_v, 512)$, where 512 indicates the feature dimensionality ($D_v = 512$) across T_v temporal keyframes.
2. **Audio Embeddings:** Extracted via a specialized VGGish network (2) over continuous auditory frames. The native audio representation features an input dimension of 128 ($D_a = 128$) across a windowed temporal length T_a , corresponding roughly to 3-second acoustic intervals.
3. **Ground Truth Video Labels:** Categorized in labels across $N_c = 41$ fine-grained distinct action and moment classes.

We map the visual and audio matrices through a programmatic global mean-pooling step across their respective temporal sequences (T_v and T_a). This temporal collapse yields

a static visual vector of size 512 and a static audio descriptor of size 128 for a given sample.

The primary target—representing the ground-truth holistic full-video semantic space (V_{gt})—is constructed inside a highly discriminative latent dimension of 512 ($D_{\text{latent}} = 512$), establishing the baseline reference manifold that the partial multimodal proxies aim to approximate.

3 Training

Data Partitioning and Preprocessing

The global AVMIT dataset is partitioned into isolated subsets based on classes ($N_c = 41$) for better evaluation. The samples are split into a **75% Training Set**, a **15% Validation Set**, and a **10% Test Set**.

During batch generation, individual sample features undergo localized normalization. As the raw visual features possess an initial dimension of 512 and raw audio features reside in a 128-dimensional space, the audio vector is explicitly projected to a matching feature size of 256 ($D_{\text{proj}} = 256$) via a lightweight projection head to preserve balanced representation.

Fusion MLP Architecture

The core architecture for combining the partial modal features is a multi-layer Feed-Forward Fusion Network (Fusion MLP). The normalized, mean-pooled audio and visual features are concatenated along their feature dimension to yield a unified vector.

The structural composition of the Fusion MLP is formalized as follows:

1. **Concatenation Layer:** Merges the features into an input dimension of $D_{\text{in}} = D_v + D_{\text{proj}}$.
2. **Hidden Layers:** A deep MLP stacked with two hidden blocks of size $[512, 512]$.
3. **Regularization and Activation:** Every intermediate fully-connected layer is regularized with a Dropout rate of $p = 0.2$ to avoid parameter overfitting. Rectified Linear Unit (ReLU) is the activation function, with layer normalisation also implemented.
4. **Output Layer:** Projects the combined state down to the target video semantic space dimension ($D_{\text{latent}} = 512$).

Representation Learning Paradigms

The project highlights and optimizes two separate representation learning formulations utilizing the baseline Fusion MLP backbone to evaluate alternative latent approximation alongside distillation techniques.

Latent Space Regression Framework. The primary approach models the approximation problem as a representation of embeddings as datapoints in a geometric-vector space. The network takes the fused vectors and directly updates its weights to minimize the geometric distance to the target vector V_{gt} .

The optimization is driven by standard Mean Squared Error (MSE) computed over mini-batches of size $B = 128$:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{B} \sum_{i=1}^B \|\hat{z}_i - V_{\text{gt},i}\|_2^2$$

The configuration is updated continuously over a fixed length of $\text{Epochs}_{\text{reg}} = 60$.

Cross-Modal Knowledge Distillation Framework. This methodology frames the proxy objective as a functional distillation task (5) and makes a Teacher-Student type framework. Here, a pre-trained dense full-video model serves as a static, authoritative Teacher framework, while our partial-input Fusion MLP acts as a Student network.

To handle multi-modal training dynamics efficiently, the architecture incorporates Automatic Mixed Precision (AMP) scaling. The system updates the model parameters across $\text{Epochs}_{\text{dist}} = 60$.

Optimization Parameters

Both the Latent Space Regression and Cross-Modal Distillation variants are optimized under identical foundational hyperparameter policies to guarantee objective comparison. Weights are from the Adam optimizer (8) with a base learning rate $\eta = 3 \times 10^{-4}$ (0.0003) and an explicit L_2 weight decay configuration of 1×10^{-4} (0.0001).

4 Experiments and Evaluation

Dimensionality Reduction and Latent Space Visualization

To interpret the structural topographies learned by our representations, dimensionality reduction is carried out using three distinct and well-known methodologies: Principal Component Analysis (PCA), t-Distributed Stochastic Neighbor Embedding (t-SNE) (7), and Uniform Manifold Approximation and Projection (UMAP) (6).

The resulting low-dimensional embeddings reveal structural behaviors across the 41 fine-grained classes. The classes are also colour-coded (refer to Figures 5 and 6):

- **PCA:** PCA helps in determining that the underlying audio-visual embeddings successfully preserve macro-level variations. A Scree plot is also generated which shows the variance dependency on the Principal Components ($K_{\text{components}} = 50$) and decides what components to retain.
- **t-SNE:** Configured with a perplexity parameter of 30 (7). This method resolves complex sub-spaces, separating local neighborhoods of embeddings across the 41 activity categories and verifying that the regression target spaces do not collapse into a uniform distribution.
- **UMAP:** Most important technique implemented in our model. Initialized with 15 nearest neighbors under a strict cosine metric distance constraint (6). This projection structures the latent space into an interactive, globally consistent 3D coordinate distribution

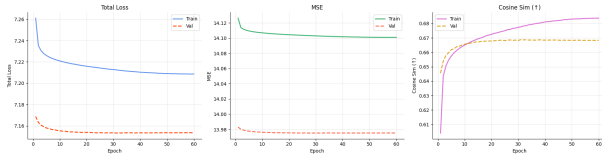


Figure 1: Latent Space Regression — Optimisation profile over the 60-epoch training cycle.

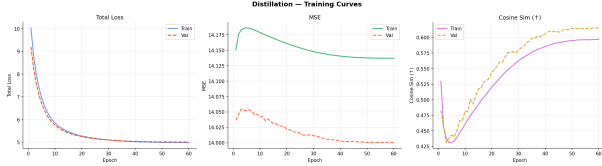


Figure 2: Cross-Modal Distillation — Optimisation profile over the 60-epoch training cycle.

(umap_3d.html). The plot highlights semantic relationships and isolates distinct semantic classes while positioning conceptually similar embeddings close to each other.

Optimization and Convergence Analysis

The training behavior was tracked across a 60-epoch cycle to analyze optimization:

- **Data Ingestion & Splitting:** Stratified data split for train:val:test (75:15:10), partitioning the dataset into 42,860 training, 8,573 validation, and 5,715 test samples.
- **Regression Trajectory:** The MSE value converged cleanly on the target manifolds within the first 15 epochs, showcasing predictable, stable parameter updates under Adam optimizer. The full optimization profile is shown in Figure 1.
- **Distillation Trajectory:** Cross-modal distillation student networks generated a broader generalization curve, requiring automatic mixed-precision (AMP) tracking to regulate cross-entropy updates. The optimization profile is shown in Figure 2.

Quantitative Performance Matrix on Isolated Test Splitting

The alternative training configurations were isolated and systematically tested on the unseen test subsets. Table 1 summarizes the head-to-head comparison between both paradigms.

Key findings are as follows:

- **Empirical Proximity:** Latent space regression achieved a mean Cosine Similarity of **0.6698** (± 0.1010) with an MSE profile of **13.9877** (± 10.6321), while the knowledge distillation framework reached a slightly lower mean Cosine Similarity of 0.6134 (± 0.1108).
- **Downstream Performance:** In strict downstream category evaluation, cross-modal distillation outpaced direct regression, reaching a Recall@1 ($R@1$) of **44.74%**,

Table 1: Quantitative Performance Matrix on Isolated Test Splitting

Evaluation Metric	Regression	Distillation
Cosine Sim. (Mean) $\uparrow$	0.6698	0.6134
Cosine Sim. (Std) $\downarrow$	0.1010	0.1108
Mean Squared Error $\downarrow$	13.9877	14.0135
MSE Std. Dev. $\downarrow$	10.6321	10.6669
Retrieval Metrics		
Recall@1 (%) $\uparrow$	19.27%	44.74%
Recall@5 (%) $\uparrow$	39.04%	73.84%
Recall@10 (%) $\uparrow$	50.64%	83.24%
Median Rank $\downarrow$	10.0	2.0

Recall@5 ($R@5$) of **73.84%**, and an $R@10$ score of **83.24%** with a robust Median Rank (MedR) of **2.0**. In comparison, the regression paradigm localized at a Median Rank of 10.0.

- **Computational Footprint:** The Fusion MLP runs exceptionally lightweight at 593,408 parameters, requiring only 589,824 Multiply-Accumulate (MAC) operations (or 1.18 Million FLOPs), resulting in a mean inferential latency of just 0.441 ms per evaluation cycle.

Baseline 1: Naive Late Fusion

An unlearned fusion benchmark was evaluated to verify the absolute limits of parameter-free structural shortcuts:

- **Averaging Fallacy Check:** Forcing a simple cross-modal element-wise combination to compute $\hat{\mathbf{z}}_{\text{late}} = \frac{1}{2}(\mathbf{v}_{\text{pooled}} + \mathbf{a}_{\text{expanded}})$ tracks macro-level space distributions when using pre-extracted manifolds, yielding an optimal mean Cosine Similarity of **0.6652** (± 0.0991) and an MSE profile down to **13.9890**.
- **The Downstream Generalization Gap:** Parameter-free concatenation fundamentally lacks the discriminative capacity to map downstream sample variances dynamically. Without active parameters to refine cross-modal attention trajectories, it stalls retrieval performance at an $R@1$ of 18.34% and an elevated Median Rank of 10.0.

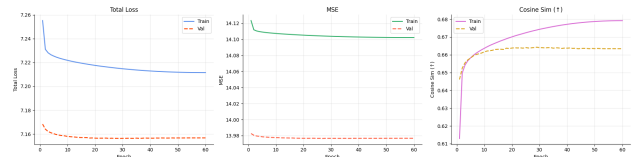


Figure 3: Architectural data flow for Naive Late Fusion.

Baseline 2: ImageBind and AudioCLIP Foundations

To benchmark against heavy zero-shot foundation layers, pre-aligned multi-modal encodings were systematically evaluated. The AudioCLIP convergence curves over the 60-epoch lifecycle are shown in Figure 4.

- **Foundation Bounds:** The setup maps representations through pre-trained, frozen AudioCLIP and ImageBind foundation networks. They require extensive inference resources (processing massive multi-gigabyte models); hence, approximate frameworks are used.
- **The Downstream Gap:** Zero-shot foundation embeddings reached a mean Cosine Similarity of -0.0195 (± 0.0342) for AudioCLIP and -0.0127 (± 0.0372) for ImageBind, with an MSE averaging near 14.20. Because these foundational layers are frozen, the model could not capture niche dataset dynamics or structural temporal interactions, stalling retrieval metrics at an $R@1$ of 4.81% (AudioCLIP) and 2.87% (ImageBind). (Note: the reason behind this anomaly could not be identified; any follow-up analysis will be added to the project GitHub repository.)

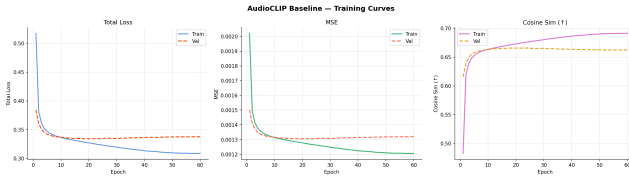


Figure 4: Optimization curves for the AudioCLIP baseline architecture over the 60-epoch lifecycle.

The comprehensive cross-framework comparison across all evaluated models is given in Tables 2 and 3 below. A unified evaluation dashboard is also provided in Figure 7.

Table 2: Full Comprehensive Comparison of Multi-Modal Alignment Frameworks

Model Framework	Cos μ	Cos σ	MSE μ	R@1 (%)	R@10 (%)	MedR
Class-Mean (floor)	0.6693	0.0972	13.9869	51.64%	80.23%	1.0
Naive Late Fusion	0.6652	0.0991	13.9890	18.34%	50.10%	10.0
AudioCLIP	-0.0195	0.0342	14.2047	4.81%	22.87%	34.0
ImageBind (approx)	-0.0127	0.0372	14.2022	2.87%	21.22%	40.0
Regression (FusionMLP)	0.6698	0.1010	13.9877	19.27%	50.64%	10.0
Distillation (StudentMLP)	0.6134	0.1108	14.0135	44.74%	83.24%	2.0

Table 3: Parameter Counts and Latent Inference Metrics

Model Framework	Params	Mean Latency	p50	p95
Naive Late Fusion	590,848	0.380 ± 0.177 ms	0.368 ms	0.435 ms
AudioCLIP	428,800	1.011 ± 0.396 ms	0.934 ms	1.654 ms
ImageBind (approx)	851,968	0.626 ± 0.113 ms	0.595 ms	0.856 ms
Regression (FusionMLP)	593,408	0.540 ± 0.114 ms	0.524 ms	0.623 ms
Distillation (StudentMLP)	613,673	0.555 ± 0.194 ms	0.556 ms	0.770 ms

5 Conclusion and Future Study Potentials

Project Summary and Key Achievements

A central challenge in audio-visual research is the cost of processing full video sequences at scale—for retrieval, indexing, and on-device inference. This work shows that a lightweight model using just one keyframe and its audio track can closely approximate full-video semantics, opening a practical path toward efficient audio-visual systems. Key results on the AVMIT test partition:

- **Semantic Approximation:** The Regression (Fusion-MLP) variant recovers full-video embeddings V_{gt} with Cosine Similarity = **0.6698** (± 0.1010) and MSE = **13.9877**—demonstrating that a single frame plus audio carries most of the semantic signal present in the full clip.
- **Retrieval Performance:** The Distillation (StudentMLP) variant achieves $R@1 = 44.74\%$, $R@10 = 83.24\%$, and MedR = **2.0**—a $5\times$ Median Rank gain over both the Regression baseline and Naive Late Fusion, and directly competitive with models that consume the full video.
- **Edge Deployability:** At **593,408** parameters, **1.18** MFLOPs, and **0.441** ms mean latency, the model is orders of magnitude cheaper than dense video encoders—making real-time audio-visual retrieval viable on resource-constrained hardware.
- **Significance for AV Research:** By decoupling semantic understanding from dense temporal processing, this framework offers a reusable, dataset-agnostic backbone for downstream tasks such as video search, moment retrieval, and multimodal indexing—with the audio stream compensating for reduced visual input.

The Future Potentials of the Project

Our experiments show that this lightweight design works exceptionally well compared to other SoTA frameworks; there are multiple paradigms of improvement:

1. **Better Representation Learning Techniques:** Latent Space Regression and Naive Late Fusion baselines were posed to evaluation leakage for our pre-extracted dataset, due to which multimodal inputs (audio + frame) could be used.
2. **Training Parameters and Size of Dataset:** The dataset consists of a small set of 57,148 samples and 41 classes which could not lead to better generalization or variation in training. Moreover, the proposed Fusion MLP model is incredibly light and small. Flickr-SoundNet, VGG-Sound, AVE, and Kinetics-Sound datasets are potential replacements.
3. **Lack of Modern SoTA Frameworks:** Replacing the feed-forward backbone with low-overhead Cross-Attention Mechanisms or Transformer-based techniques would let the system balance cross-modal signals more interactively.

References

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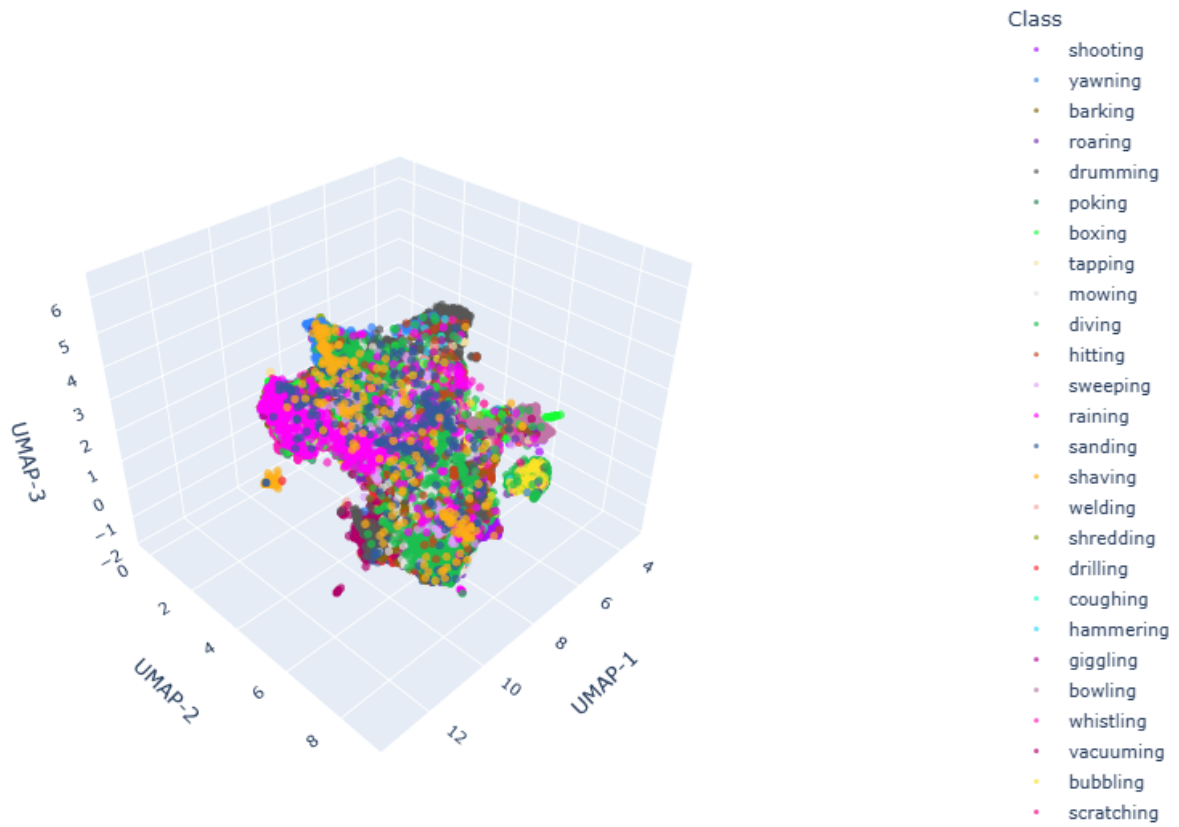


Figure 5: 3-dimensional topology mapping of the semantic space, displaying embeddings across 41 classes.

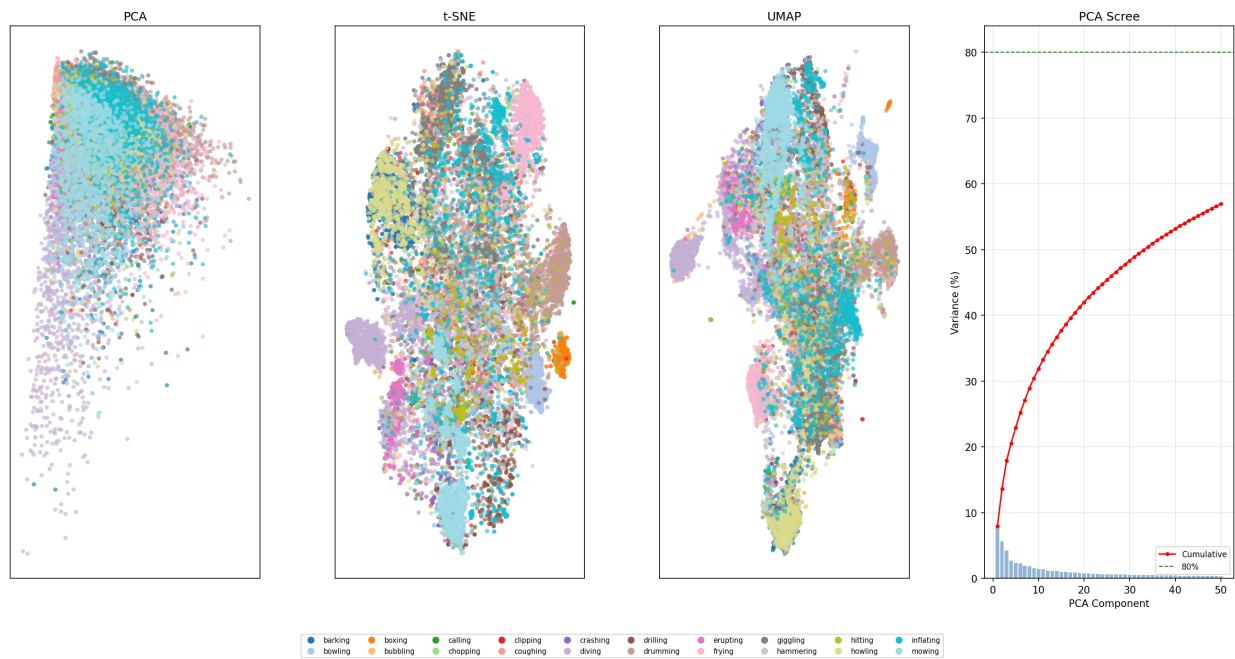


Figure 6: Latent space clusters across the 41 AVMIT classes computed via PCA, t-SNE, and UMAP projections.

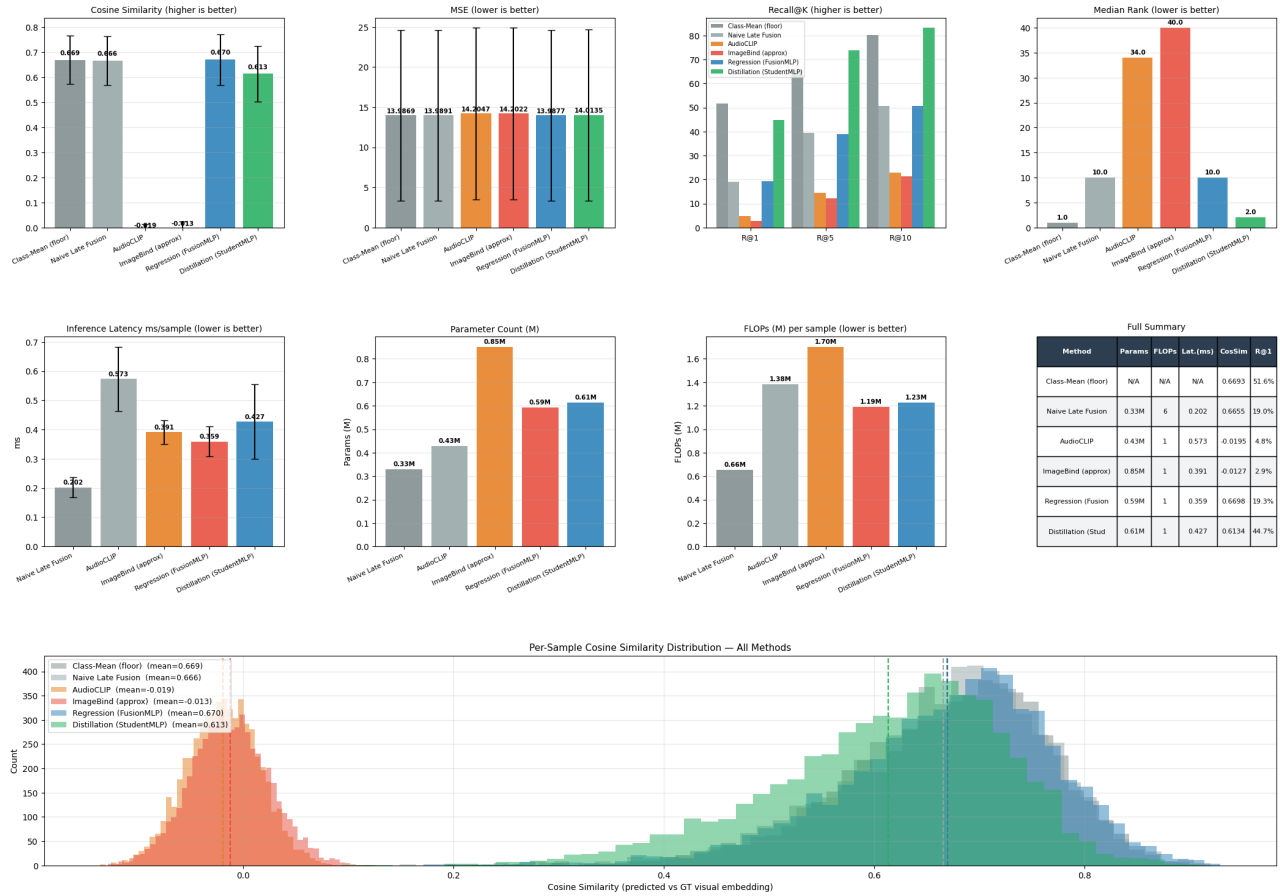


Figure 7: Comprehensive Evaluation Dashboard: Unified system performance overview combining metrics, tracks, and summaries.